

Hodge groups, exceptional classes, and odd-exponent abelian covers

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5 September 2026

Abstract

We prove a transfer criterion for full branch families of abelian covers of $\mathbf{P}^1$ of any odd exponent N : the rational Hodge conjecture on every power of the degree- N Fermat Jacobian implies it on every power of the very general covering Jacobian. Aoki’s theorem makes this unconditional for odd prime powers and for $N = 3^a 5^b$ or $3^a 7^b$. The proof includes the exceptional constant primitive factor at conductor fifteen and its algebraic elliptic realization. For prime-exponent families, actual oriented graph correspondences determine a rational ambient group whose derived subgroup is connected monodromy; a signed determinant lattice then gives the exact generic Hodge group. Branch words formed from an odd multiplicative subgroup and opposite pairs yield moving absolutely simple Jacobians with an explicitly counted first exceptional Hodge quotient. These classes are algebraic already on the Jacobian itself and generate all additional Hodge classes on its powers under homomorphism pullbacks. The Hodge rings of distinct-prime parts factor even when their covers share the full branch configuration. Finally, the degree-seven order-three family satisfies the generalized Hodge conjecture on every power, as an explicit corollary of classical cyclic monodromy, Fermat-cycle algebraicity, and Abdulali’s domination theorem. These are written arguments depending on the stated companion results and standard published inputs. They are not complete Lean formalizations, independently refereed results, or claims of priority.

1 Scope and dependencies

The companion [1] treats a full ordered branch family of connected elementary covers of $\mathbf{P}^1$ of fixed prime exponent. It proves the connected monodromy blocks, constructs actual oriented graph correspondences, and realizes their invariant tensors through rational algebraic maps from determinant-word Hodge structures. The companion [3, Corollary 1.2] transfers these determinants to prime-degree Fermat varieties and applies Shioda’s published product theorem. It proves algebraicity of every rational Hodge class in their tensor products, including all-row-balanced mixed determinants with repeated curves and different chosen generators. The compact-type fusion argument [2] supplies an alternative route for the balanced subspaces. We use these specific results, not the frozen earlier manuscripts.

Section 7 supplies the additional primitive-factor and constant-factor arguments needed at odd composite exponents. Its cycle premise is the all-powers conjecture for one Fermat Jacobian, verified in the stated cases by [15]; the Shioda–Katsura construction [16] transfers that premise to the required Fermat products. The earlier sections retain their stated prime-exponent scope.

Write $\text{Hdg}^d(A) = H^{2d}(A, \mathbf{Q}) \cap H^{d,d}(A)$, and let $\text{Div}^d(A)$ be the degree- $2d$ part of the rational algebra generated by divisors. “Exceptional” below means outside that algebra. All assertions concern rational cohomology over $\mathbf{C}$. Very general always refers to the indicated full configuration space, not an arbitrary restricted subfamily.

2 The exact generic Hodge group

Fix an odd prime p , $K = \mathbf{Q}(\zeta_p)$, and $\Gamma = \mathbf{F}_p^\times = \text{Gal}(K/\mathbf{Q})$. Let $G = (\mathbf{Z}/p)^m$ and let $a_1, \dots, a_r \in G \setminus \{0\}$ generate G with sum zero. At a very general point of the full ordered branch family put $A = J(X)$ and $V = H^1(A, \mathbf{Q})$. Use the evaluation code

$$C = \{(c(a_i))_i : c \in \text{Hom}(G, \mathbf{F}_p)\} \subset \mathbf{F}_p^r.$$

For a nonzero word c , put $s(c) = |\{i : c_i \neq 0\}|$ and $h(c) = s(c) - 2$. Discard support-two factors, which have genus zero. Over K write V_c for the character space; its dimension is $h(c)$. All spaces retained have support at least three.

Define oriented classes as follows. On a support of size four, c and e are equivalent when $e = Pc$ for a labelled Klein-four double transposition or the identity, with both words in C . On other supports the classes are singletons. Write O_c for the class. Negation acts on these classes. By [1, Section 3], each oriented relation comes from an actual rational algebraic graph map on the cyclic factors, after one common finite base cover. A self-negative class occurs only in rank two, has opposite-paired residues, and has signature $(1, 1)$ at every embedding.

Let $\mathcal{E} \subset \text{End}_{\mathbf{Q}}(V)$ be the algebra generated by the deck action, the oriented graph maps transported to A by quotient pullback and norm, and their polarization adjoints. Define

$$L = \text{Cent}_{\text{Sp}(V)}(\mathcal{E})^0, \quad M = \text{connected algebraic monodromy on } V.$$

Every generator of $\mathcal{E}$ is an algebraic Hodge endomorphism, and commutes with monodromy after the stated finite cover.

Lemma 2.1. *The group L is split over K , and*

$$L_K \simeq \prod_{\{O, -O\}, O \neq -O} \text{GL}_{h(O), K} \times \prod_{O=-O} \text{SL}_{2, K}.$$

A non-self-negative pair contributes standard occurrences on O and dual occurrences on $-O$. Support-three classes contribute GL_1 factors. Moreover $L^{\text{der}} = M$ as subgroups of $\text{Sp}(V)$.

Proof. The character idempotents

$$e_c = \frac{1}{|G|} \sum_{g \in G} \zeta_p^{-c(g)} g^* \in K[G]$$

are defined over K . Commuting with the deck action therefore forces preservation of every V_c already over K . The polarization pairs V_c nondegenerately with V_{-c} . If ϕ is an actual rational oriented graph map, its component $e_e \phi e_c$ is a K -linear isomorphism $V_c \rightarrow V_e$. Centralizing it requires conjugate actions on these two spaces. Different paths between the same spaces differ by a scalar: they commute with the irreducible standard monodromy action; this can first be checked over $\mathbf{C}$ and descended to K . Thus additional paths impose no further equations on a chosen matrix.

For $O \neq -O$, the free matrix on a representative space is an arbitrary $\text{GL}_h(K)$ matrix; all other matrices are obtained by the graph maps and polarized duality. In a self-negative class, an actual map $V_c \rightarrow V_{-c}$ combined with the polarization gives a nondegenerate monodromy-invariant bilinear form on V_c . The standard SL_2 representation has its alternating form as its unique invariant bilinear form, up to a scalar. The resulting form is therefore alternating over K , and its preserving group is the split $\text{Sp}_2 = \text{SL}_2$.

This calculates the centralizer over K , not merely its complexification; all displayed factors are connected. Its derived subgroup is precisely the product of the moving SL_h blocks in [1, Theorem 3.1]. Support-three factors have trivial connected monodromy. Hence its derived subgroup and M are equal inside the original faithful representation. $\square$

Put $D = L/M$ and let $q : L \rightarrow D$ be the quotient. Choose one class from each non-self-negative pair. Its determinant character is denoted e_O ; then

$$E = X^*(D_K) = \bigoplus_{\{O, -O\}, O \neq -O} \mathbf{Z}e_O.$$

Reversing orientation changes e_O to $-e_O$. A self-negative class contributes no generator; retaining a formal relation $2e_O = 0$ would give an incorrect unsaturated presentation. The determinant on each GL_h factor has kernel the actual SL_h , so no h th power belongs in this quotient map. This is the quotient-torus lattice, rather than an identification with the lattice of an isogenous central torus.

Galois acts on E by the signed permutation of oriented labels. The K -rational calculation in Lemma 2.1 justifies that these are the complete descent data on D : changes of graph paths or bases act by inner conjugation and leave determinants unchanged.

For a representative $c \in O$, set

$$d_c(b) = s(c) - 1 - \frac{1}{p} \sum_{i:c_i \neq 0} [bc_i]_p, \quad \delta_c(b) = 2d_c(b) - h(c),$$

where $[x]_p \in \{1, \dots, p-1\}$. Oriented graphs preserve d , negation changes the sign of δ , and $\delta = 0$ on self-negative classes. Define the integral map

$$\Sigma : E \longrightarrow \mathbf{Z}^\Gamma, \quad e_O \longmapsto (\delta_c(b))_{b \in \Gamma}.$$

Its kernel is saturated. Let $T \subset D$ be the subtorus with annihilator $\ker \Sigma$, so that

$$X^*(T_K) = E / \ker \Sigma \simeq \mathrm{im} \Sigma.$$

The quotient is primitive; this does not claim that $\mathrm{im} \Sigma$ is saturated in the unrelated ambient lattice $\mathbf{Z}^\Gamma$.

Theorem 2.2. *At a very general point of this full branch family,*

$$\mathrm{Hg}(A) = q^{-1}(T) \subset L.$$

Consequently

$$\dim \mathrm{Hg}(A) = \sum_{\text{moving complex blocks}} (h^2 - 1) + \mathrm{rank}_{\mathbf{Q}} \Sigma, \quad \mathrm{MT}(A) = \mathbf{G}_m \cdot \mathrm{Hg}(A) \subset \mathrm{GL}(V).$$

The sum counts each complex simple block once, including self-negative rank-two blocks.

Proof. The generic monodromy inclusion gives $M \subset \mathrm{Hg}(A)$, while the algebraic Hodge endomorphisms defining L give $\mathrm{Hg}(A) \subset L$. The projected Hodge circle in D pairs with e_O by the integer $2d_c(1) - h(c)$. The smallest rational subtorus containing this circle is obtained by adjoining all its Galois-conjugate cocharacters. Since D splits over K , their pairings are exactly the rows of Σ . Its annihilator is therefore $\ker \Sigma$, and this subtorus is T .

By minimality of the Hodge group, $\mathrm{Hg}(A) \subset q^{-1}(T)$; its image under q contains T and is contained in T . Since it contains the full kernel M , it equals $q^{-1}(T)$. This preimage is connected because both its kernel and quotient are connected. The dimension and weight-one Mumford–Tate formulas follow. The last product is not asserted to be a direct product. $\square$

Remark 2.3. The minimal rational span of a Hodge-signature vector is standard: compare Xue–Zarhin [5, Lemma 3.9, Theorem 3.11 and Section 3.13]. The geometric work here is the exact ambient centralizer with its actual oriented maps, including self-negative rank-two and constant rank-one sectors. An unproved replacement of Hodge endomorphisms by the entire monodromy commutant would not establish this theorem.

3 Exceptional classes on a single simple Jacobian

Let $P \subset \mathbf{F}_p^\times$ be a multiplicative subgroup of odd order $m \geq 3$. Choose $k \geq 0$ with $s = m + 2k \geq 5$, and take the ordered branch word

$$a = (\text{one occurrence of every element of } P, \underbrace{1, -1, \dots, 1, -1}_{k \text{ pairs}}). \quad (1)$$

Its sum is zero, because the elements of P are the roots of $x^m - 1$. Let $Y \rightarrow \mathbf{P}^1$ be the associated cyclic cover and $A = J(Y)$. Put

$$h = s - 2 = m + 2k - 2 \geq 3, \quad g = \frac{(p-1)h}{2}, \quad r_0 = (p-1)/2.$$

The integer h is odd. Write $V_b \subset H^1(A, \mathbf{C})$ for the deck eigenspace and $D_b = \bigwedge^h V_b$ for its volume line.

Theorem 3.1. *For every smooth member of this family there is an algebraic rational subspace $\mathcal{E} \subset \text{Hdg}^h(A)$ with*

$$\mathcal{E}_{\mathbf{C}} = \bigoplus_{\substack{\{b, -vb\} \text{ distinct unordered pairs} \\ b \in \mathbf{F}_p^\times, v \in P \setminus \{1\}}} D_b \wedge D_{-vb}, \quad \dim_{\mathbf{Q}} \mathcal{E} = \frac{(p-1)(m-1)}{2}. \quad (2)$$

At a very general member, $\text{End}^0(A) = K$, the Jacobian is absolutely simple and not of CM type, and

$$\text{Hdg}^d(A) = \text{Div}^d(A) \quad (d < h), \quad \text{Hdg}^h(A) = \text{Div}^h(A) \oplus \mathcal{E}.$$

In particular h is its first exceptional codimension, and

$$\dim_{\mathbf{Q}} \text{Hdg}^h(A) = \binom{r_0 + h - 1}{h} + \frac{(p-1)(m-1)}{2}.$$

Algebraicity and diagonal realization. Fix $v \in P \setminus \{1\}$. Use Y twice, with second generator $\sigma_2 = \sigma^{-v^{-1}}$. A generator change $\sigma \mapsto \sigma^e$ changes residues by $a_i \mapsto e^{-1}a_i$. Thus the second residue word is $-va$, and its b -eigenspace is the original V_{-vb} . The concatenation $a \sqcup (-va)$ is a union of opposite pairs: its subgroup portion is $P \sqcup (-P)$ because $vP = P$, and its added pairs remain opposite pairs. The cyclic Hodge-dimension formula therefore makes the following rational space entirely of Hodge type (h, h) :

$$(W_v)_{\mathbf{C}} = \bigoplus_{b \neq 0} D_b \otimes D_{-vb} \subset H^h(A) \otimes H^h(A)$$

It has dimension $p-1$ and is algebraic by the Fermat transfer [3, Corollary 1.2]. Equivalently one can use balanced fusion [2, Theorem 1.1].

Pull back along $\Delta : A \rightarrow A \times A$. Since P has odd order, $-1 \notin P$ and $b \neq -vb$; cup multiplication is nonzero on every displayed line. Equality $\{b, -vb\} = \{b', -vb'\}$ either gives $b = b'$ or, by exchanging endpoints, $v^2 = 1$. The latter is impossible for $v \neq 1$ in P . Hence Δ^* is injective on W_v , and its image $\mathcal{E}_v$ has dimension $p - 1$.

For different parameters, endpoint exchange shows that their unordered label pairs coincide exactly when the parameters are v and v^{-1} . Thus $\mathcal{E}_v = \mathcal{E}_{v^{-1}}$, and otherwise their exterior multi-degrees are disjoint. Summing over the $(m - 1)/2$ inverse pairs proves (2). Rationality and algebraicity follow from the rational determinant spaces and algebraic diagonal pullback. $\square$

Simplicity and the complete first quotient. For this cyclic family $s \geq 5$, the connected monodromy theorem gives

$$M_{\mathbf{C}} = \prod_{\{b, -b\}} \mathrm{SL}_h,$$

standard on V_b and dual on V_{-b} . Since $h \geq 3$, these two representations are inequivalent. Its commutant on $H^1(A, \mathbf{C})$ consists of a scalar on each of the $p - 1$ eigenspaces. The known rational algebra K has this full dimension, so $\mathrm{End}_M(H^1(A, \mathbf{Q})) = K$. At a very general point every Hodge endomorphism commutes with M ; the algebraic deck action then gives $\mathrm{End}^0(A) = K$. A proper abelian subvariety would give a nontrivial rational idempotent by Poincaré reducibility, impossible in this field. Also $[K : \mathbf{Q}] = p - 1 < 2g$, so A is not CM.

For one block $U \oplus U^*$ with $\dim U = h$, the exterior invariants of SL_h are generated by a polarization tensor θ of degree two and the two volume lines of degree h . Indeed $\bigwedge^i U$ is irreducible for $0 < i < h$, and

$$(\bigwedge^i U \otimes \bigwedge^j U^*)^{\mathrm{SL}_h}$$

is one-dimensional for $i = j$ and otherwise zero, apart from the trivial endpoint exterior powers for $i, j \in \{0, h\}$. The paired invariants are the nonzero powers θ^i , $0 \leq i \leq h$, and the product of the two volumes is a nonzero multiple of θ^h .

Invariants for the product group tensor together. In even degree below $2h$, the oddness of h prevents a single volume, while two volumes have too large a degree. Only products of pairings remain. The degree-two invariant space consists entirely of divisors: equivalently

$$\mathrm{NS}(A)_{\mathbf{Q}} \simeq \mathrm{End}^0(A)^{\dagger=1} = K^+,$$

and its complexification contains each of the r_0 pairing lines. Thus all Hodge classes below codimension h are divisor-generated.

In degree $2h$, the additional invariants are pairs of volumes. Opposite labels already give powers of a pairing, and equal labels give zero in the exterior algebra. The remaining rational Galois orbits are $\mathcal{E}_v$ for $v \neq \pm 1$, with v identified with v^{-1} . The diagonal argument above gives each such orbit a rank-one K -Hodge structure of dimension $p - 1$, regardless of whether it is Hodge. Its rational Hodge subspace is zero or the whole space. Writing $d_a(b) = \dim V_b^{1,0}$, it is the whole space exactly when

$$d_a(vb) = d_a(b) \quad \text{for every } b \neq 0. \tag{3}$$

The stabilizer in (3) is exactly P . Inclusion of P follows from its multiplicative invariance and the constant contribution of every opposite pair to the fractional-part sum. Conversely (3) is all-row balance for $a \sqcup (-va)$. The prime balanced-tuple equality of Aoki–Parry, in the precise form recorded in [2, Section 3], forces its residue multiplicities to be invariant under negation. Cancelling the added pairs gives

$$1_P - 1_{-P} = 1_{vP} - 1_{-vP}.$$

Distinct multiplicative cosets are disjoint and $P \neq -P$, so this equality forces $vP = P$. Thus precisely the spaces in (2) supply the additional Hodge classes.

Divisor products have equal occupation numbers in V_b and V_{-b} for every label pair. A line $D_b \wedge D_{-vb}$ with $v \neq \pm 1$ does not. Their exterior multi-degrees are disjoint, proving $\mathcal{E} \cap \text{Div}^h(A) = 0$. The pairing algebra is

$$\mathbf{C}[\theta_1, \dots, \theta_{r_0}]/(\theta_i^{h+1}).$$

In degree h in these generators the truncations remove no monomial, so its dimension is $\binom{r_0+h-1}{h}$. This proves the assertion. $\square$

Remark 3.2 (Moving and unbounded examples). The parameter space has dimension $s - 3 > 0$. A fixed curve of genus at least two has finitely many cyclic actions of order p , so forgetting the covering structure has finite fibers; Torelli gives the same dimension for the Jacobian image.

For every prime $p \equiv 1 \pmod{3}$, choose an order-three element u and take $P = \{1, u, u^2\}$, $k \geq 1$. Then $h = 2k + 1$, $g = (p - 1)(2k + 1)/2$, and $\dim \mathcal{E} = p - 1$. At $p = 7$, the word $(1, 2, 4, 1, 6)$ gives a two-dimensional family of absolutely simple ninefolds with

$$\dim \text{Hdg}^3(A) = 10 + 6 = 16.$$

The six exceptional lines have unordered labels $\{1, 5\}, \{2, 3\}, \{1, 3\}, \{4, 6\}, \{4, 5\}, \{2, 6\}$.

Alternatively, for $p \equiv 3 \pmod{4}$, $p \geq 11$, use the quadratic residues and $k = 0$. Then

$$h = (p - 5)/2, \quad g = (p - 1)(p - 5)/4, \quad \dim \mathcal{E} = (p - 1)(p - 3)/4.$$

The word $(1, 3, 4, 5, 9)$ at $p = 11$ gives twenty exceptional algebraic codimension-three classes on a moving simple fifteenfold. These formulas give unbounded prime degrees, dimensions and codimensions.

4 Generators on every power of the subgroup family

The first exceptional space suffices on all powers, provided it is allowed to move between copies by homomorphism pullback.

Theorem 4.1. *For a very general A in (1), and every $q \geq 1$, the rational Hodge ring of A^q is generated by its divisors and*

$$f^* \mathcal{E} \subset \text{Hdg}^h(A^q), \quad f \in \text{Hom}(A^q, A).$$

All these generators are algebraic. No exceptional Hodge class occurs on any power in codimension strictly less than h .

Proof. The rational diagram and Hodge-lifting proof of [1, Sections 5–6] expresses every Hodge class as a span of pairing diagrams and all-row-balanced effective volume words. For this cyclic family an effective word is a list of labels $b_1, \dots, b_N$; its branch words are the scalar multiples $b_j a$. The added opposite pairs make no contribution to antisymmetric residue multiplicities. Aoki–Parry therefore says that balance is equivalent to

$$\sum_j (1_{b_j P} - 1_{-b_j P}) = 0.$$

As $P \neq -P$, this means that each coset and its negative occur equally often. Pair the volume occurrences accordingly. Each pair has labels $b, -vb$ for some $v \in P$. This pairing of a word is

compatible with its full Galois orbit: simultaneous multiplication of all labels preserves each chosen value of v . No negative multiplicities are introduced.

For $v = 1$, the paired volume tensors are alternating products of standard–dual coevaluations; hence their realizations are generated by divisors. For $v \neq 1$ they are the volume pairs defining $\mathcal{E}$. It remains to show that homomorphism pullbacks realize all their possible placements in the copies of A^q , rather than just placing both volumes in one copy.

For a label b , the standard space in $H^1(A^q, \mathbf{C})$ is $V_b \otimes \mathbf{C}^q$. Its alternating h -volume subspace is

$$D_b \otimes \mathrm{Sym}^h(\mathbf{C}^q).$$

This follows either from the exterior Cauchy formula or by alternating the V_b slots: because their determinant is alternating, the copy indices are symmetric. Pullback by $f = \sum_i a_i \mathrm{pr}_i$, $a_i \in K$, sends D_b to its tensor with the pure h th power of the coefficient vector $(\tau_b(a_i))_i$. For a nonopposite pair $b, c = -vb$, the vectors at b and c vary independently after taking complex linear spans. Indeed K^q is Zariski dense in the affine $\mathbf{Q}$ -space $\mathrm{Res}_{K/\mathbf{Q}} \mathbb{A}^q$: choose a $\mathbf{Q}$ -basis of K and use the density of $\mathbf{Q}^{q(p-1)}$. After extension to $\mathbf{C}$ the embedding coefficient vectors are independent coordinates. Pure powers span symmetric powers in characteristic zero. Thus these pullbacks span

$$(D_b \wedge D_c) \otimes \mathrm{Sym}^h(\mathbf{C}^q) \otimes \mathrm{Sym}^h(\mathbf{C}^q),$$

including every placement of this pair of volumes.

Rational coefficients in the a_i cause no problem: clearing a common denominator gives an actual homomorphism, and its action in degree $2h$ differs by a nonzero rational scalar. Likewise use a rational basis of $\mathcal{E}$ before extending scalars; no single complex line is assumed to be a rational generator. Cup products of the resulting pair-volume classes and divisors span each balanced diagram in its prescribed copy positions. Faithful scalar extension then gives the asserted rational generation. Every generator is algebraic by Theorem 3.1 and algebraic pullback. Finally a tensor invariant of even degree below $2h$ cannot contain an odd number of degree- h volumes, nor two such volumes; only the divisor pairings remain. $\square$

5 Products belonging to different primes

The distinction between separate validity of the Hodge conjecture and its validity for a product requires an additional argument. Here independent cyclotomic embeddings provide it, including for constant CM factors.

Lemma 5.1. *Let $K_1, \dots, K_a$ be jointly linearly disjoint finite Galois CM fields. Let W_i be a pure rational Hodge structure of weight n_i , rank one over K_i , with its field action by Hodge endomorphisms. Suppose $\sum n_i + 2t = 2d$. Then*

$$\bigotimes_i W_i \otimes \mathbf{Q}(-t)$$

has a nonzero rational Hodge class if and only if every n_i is even and W_i is entirely of type $(n_i/2, n_i/2)$. In that event its Hodge vectors are spanned by external tensors of individual Hodge vectors and the Tate vector.

Proof. The tensor product $K_1 \otimes \dots \otimes K_a$ is the compositum field, and the displayed source has rank one over it. Its rational Hodge subspace is field-stable, so is zero or the whole source. The embeddings of this field are independent tuples of embeddings of the K_i . If $f_i(\tau)$ is the first Hodge

index on the τ -line of W_i , all tuple sums $\sum f_i(\tau_i) + t$ must therefore equal d . Varying one embedding makes each f_i constant. Conjugation gives $f_i(\bar{\tau}) = n_i - f_i(\tau)$, forcing the constant to be $n_i/2$. The converse and the tensor-span assertion follow immediately. $\square$

For distinct odd primes, the fields $K_p = \mathbf{Q}(\zeta_p)$ are jointly linearly disjoint: their compositum is $\mathbf{Q}(\zeta_{\prod p})$ and its degree is $\prod(p-1)$, with Galois group $\prod \mathbf{F}_p^\times$, by the cyclotomic degree theorem and the Chinese remainder theorem. The binary part has only Tate invariant sources.

Theorem 5.2. *For each prime in a finite set, prescribe a connected elementary cover of $\mathbf{P}^1$ with that prime exponent. Either let their full ordered branch configurations vary independently, or use one common full ordered configuration, allowing arbitrary overlaps among active labels. At a very general parameter, write A_p for their Jacobians. For every choice of nonnegative powers q_p , external product gives an isomorphism*

$$\bigotimes_p \mathrm{Hdg}^*(A_p^{q_p}) \simeq \mathrm{Hdg}^*\left(\prod_p A_p^{q_p}\right).$$

Every class on the right is algebraic. Each A_p includes its constant CM part. The assertion also passes to abelian isogeny factors of these prime parts.

Independence of the moving groups. For independently varying configurations, the fundamental group of the product parameter space is the product and the monodromy acts in separate summands. Its connected closure is the product of the individual groups. Consider a common full configuration instead. Forgetting inactive labels has connected configuration-space fibers, giving full projection on each individual monodromy group. The connected group is semisimple: its radical projects trivially to every semisimple moving projection and acts trivially on the constant factors. A proper subdirect product would, by semisimple Goursat, identify two isomorphic simple adjoint blocks.

Fix an odd-prime moving character of prime p . There is an active pair i, j whose residue sum is nonzero; otherwise three active nonzero entries would have all pair sums zero, contradicting that p is odd. The labelled pair twist has eigenvalues $(\zeta_p^{c_i+c_j}, 1, \dots, 1)$ projectively, and hence exact projective order p . On a different odd-prime ℓ character, the same twist is either identity (an inactive label), unipotent (zero pair sum), or has projective order ℓ . These are the pair formulas and nonzero-rank-one checks in [1, Section 3], from Menet–Nguyen [4, Theorem 2.5].

This comparison is legitimate before finite auxiliary base changes. The projective representations are defined on the original common marked-sphere fundamental group. A graph between two centerless simple groups is its own normalizer in their product: a normalizing pair differs from a graph pair by an element centralizing the target group. Since connected closure is normal in full closure, any connected graph relation would hold on every original pair twist. An algebraic group automorphism preserves element order, so the orders just listed exclude such a graph across distinct odd primes.

An odd-prime SL_h and binary Sp_{2g} can have isomorphic simple groups only for $h = 2, g = 1$: equal rank and dimension give $h-1 = g$ and $h^2-1 = g(2g+1)$, hence $g(g-1) = 0$. In that case the same binary twist is identity or a Picard–Lefschetz unipotent, of infinite order when nonidentity; it cannot match the odd-prime finite order. Thus all cross-prime graphs are excluded. The connected monodromy is exactly the product of the prime-part groups. $\square$

Hodge-source separation and algebraicity. In every cohomological degree n , the rational diagram construction of [1, Section 5] gives an algebraic rational Hodge surjection onto $H^n(A_p^{q_p}, \mathbf{Q})^{M_p}$ from

a finite sum of sources

$$W_{p,w} \otimes \mathbf{Q}(-t), \quad H(w) + 2t = n,$$

and Tate sources. Each nonempty word source is rank one over K_p . Although the companion's final application sets $n = 2d$, the diagram construction itself works in arbitrary degree: none of its slot, alternation, Galois-orbit or graph operations requires even n . This observation is necessary for the odd/odd Künneth components here. For $p = 2$ the invariant sources are entirely Tate.

In each Künneth multidegree, independence of the moving groups makes the invariant target the tensor product of its individual invariant spaces. Tensor their source maps. The resulting sources are finite sums of tensor products with at most one rank-one K_p word for each odd prime, together with Tate twists. Lemma 5.1 says that a nonzero Hodge vector in such a source forces every individual prime-word source to be balanced and of even weight. In particular odd prime-part cohomological degrees supply no Hodge classes in the global decomposition.

At a very general point, Hodge classes are fixed by connected monodromy. Lift each through this rational Hodge surjection using semisimplicity of polarizable pure Hodge structures. Its components are Hodge separately in the displayed rational source summands, so they are spanned by external products of individual balanced Hodge vectors. The forward maps are algebraic and each balanced word is algebraic by [3, Corollary 1.2], or alternatively by same-prime fusion. This proves algebraicity and external-product surjectivity. Künneth gives injectivity and compatibility with multiplication.

No algebraicity of the Hodge splitting is needed. Support-three constant factors are retained as rank-one determinant entries throughout. An abelian isogeny factor is reached by rational algebraic inclusion and retraction; pulling back and then retracting classes proves the same assertions for its powers. Prime-part algebraic projectors also preserve the stated factorization. $\square$

Remark 5.3 (Product boundary). Theorem 5.2 concerns products of prime-cover Jacobians and their isogeny factors; the entire Jacobian of a composite-exponent compositum has additional mixed-order characters. The separate criterion in Section 7 treats those full Jacobians at its stated odd exponents. Neither assertion replaces the full configuration by a subfamily. Jointly very general parameters are required; separate genericity alone is not substituted. The special Hodge groups can equivalently be recovered as a product from the factorization of all tensor invariants, but the full Mumford–Tate groups share their weight direction and must not be described as an unrestricted direct product.

6 A classical generalized-Hodge corollary in degree seven

Here the stronger conjecture follows from an additional domination argument, not from ordinary all-powers algebraicity alone. We give the argument because the auxiliary CM varieties can be identified explicitly. The ordinary-Hodge input in this section uses only a single moving cyclic cover and a fixed CM elliptic curve; the simultaneous elementary-abelian theorem is unnecessary.

Theorem 6.1. *For $k \geq 1$, let A be the Jacobian at a very general point of the full ordered cyclic degree-seven branch family with word*

$$(1, 2, 4, (1, -1)^k).$$

Then $\dim A = 3(2k + 1)$, and the generalized Hodge conjecture holds for every power of A .

Signatures and the domination hypothesis. Put $h = 2k + 1$, $K = \mathbf{Q}(\zeta_7)$, $V = H^1(A, \mathbf{Q})$, and $P = \{1, 2, 4\} \subset \mathbf{F}_7^\times$. Rohde's prime-cyclic monodromy theorem [10, Theorem 5.3.7] gives $M_{\mathbf{C}} = \mathrm{SL}_h^3$.

As in Section 3, its commutant is K , so $\text{End}^0(A) = K$ and A is absolutely simple of Albert type IV. The inclusions $M \subset \text{Hg}(A) \subset L(A)$, where $L(A)$ is the Lefschetz group, give

$$\text{Hg}(A)^{\text{der}} = L(A)^{\text{der}} = M.$$

This is precisely the definition of pel-type in [13, Section 2]; it does not assert that the branch family fills a PEL moduli space.

Multiplication by $b \in P$ permutes $(1, 2, 4)$, whereas its negative has least-residue sum fourteen. Each added opposite pair has sum seven. Thus the Hodge multiplicities of V_b are $(k+1, k)$ for $b \in P$ and $(k, k+1)$ for $b \in -P$. Every real unitary signature differs by exactly one. Abdulali's Theorem 10 therefore makes A weakly self-dominated, and his Lemma 9 gives the same property for every A^q . The difference-one condition is essential: it is also the correction to the older type-IV statements in [14, Appendix B.5]. We use the 2012 formulation with that condition explicitly imposed. $\square$

Identification of every required CM partner. Let $D = \bigwedge_K^h V$ denote the rational determinant realization. Its twist $D(k)$ has weight one, with type $(1, 0)$ on P and $(0, 1)$ on $-P$. The three-point curve

$$C_0 : y^7 = x(x-1)^2$$

has these same K -types. Both spaces have rank one over K , so $D(k) \simeq H^1(C_0, \mathbf{Q})$ as K -linear Hodge structures.

The fixed field K^P is $K_0 = \mathbf{Q}(\sqrt{-7})$. Indeed, $z = \zeta_7 + \zeta_7^2 + \zeta_7^4$ satisfies $z + \bar{z} = -1$ and $z\bar{z} = 2$, hence $z^2 + z + 2 = 0$. Choose a CM elliptic curve E with field K_0 and the type whose extensions to K are P . Its induced K -Hodge structure has the preceding types, whence

$$D \simeq H^1(E, \mathbf{Q})^{\oplus 3}(-k), \quad J(C_0) \sim E^3.$$

Here the isogeny follows from the equivalence of rational weight-one Hodge structures and complex abelian varieties up to isogeny.

It remains to handle the entire determinant tensor algebra, not just D . The Hodge group of E is the quadratic norm-one torus. For $n \geq 1$ its rational character pair $\chi^n \oplus \chi^{-n}$ has a realization U_n of types $(n, 0), (0, n)$ in $H^n(E^n, \mathbf{Q})$: take the rational sum of the all-holomorphic and all-antiholomorphic lines in $H^1(E)^{\otimes n}$. The two lines are Galois conjugate, so their sum is rational. Put $U_0 = \mathbf{Q} \subset H^0(E, \mathbf{Q})$. Every irreducible constituent of $D^{\otimes N}$ is a copy of

$$U_n \left(-Nk - \frac{N-n}{2} \right), \quad 0 \leq n \leq N, \quad n \equiv N \pmod{2}.$$

Twisting by $Nk + (N-n)/2$ realizes it as U_n in a power of E . These realizations are fully twisted: they are effective and have a nonzero form of type $(n, 0)$. Trivial constituents use U_0 ; duals and Tate factors change only the twist, not the CM character.

In the proof of [13, Theorem 11], the auxiliary CM category is the tensor algebra of the determinants of the simple isogeny factors. The only simple factor of A^q is A , so the object just computed is exactly the required one. By [13, Remark 12], every A^q is consequently dominated by varieties $A^n \times E^t$: each irreducible Hodge structure in its cohomology has a fully twisted Tate-equivalent realization on one of these varieties. No assertion about arbitrary CM types over K is involved. $\square$

Ordinary Hodge conjecture on the required products. We verify ordinary HC for all $A^u \times E^v$ directly from the classical cyclic and Fermat inputs. The fixed factor E has trivial monodromy over the

branch configuration, so connected monodromy of the product remains M , acting only on V . Special-linear invariant theory expresses all M -invariant tensors through polarization contractions, full determinant tensors, and unrestricted tensors in $H^1(E)$. Taking rational Galois orbits of the diagrams gives a rational Hodge surjection, in each degree, from a finite direct sum of tensor products of copies of D , $H^1(E)$, and Tate factors onto the monodromy invariants. Its maps are algebraic correspondences: deck projectors, permutations, alternation, polarization, and the usual cup-product correspondences. This is the single-cyclic instance of the explicit construction in [1, Section 5]; it requires no comparison of distinct moving cyclic quotient lines.

The determinant D is an algebraic direct summand of the middle cohomology of a degree-seven Fermat h -fold, by the classical finite-quotient construction written out in [3, Theorem 1.1]. The curve C_0 is a quotient of the degree-seven Fermat curve: in affine coordinates $r^7 + s^7 = 1$, set $x = r^7$ and $y = rs^2$. Then $y^7 = x(x-1)^2$. Since E is an isogeny factor of $J(C_0)$, its first cohomology is an algebraic direct summand of that of the Fermat curve. Thus every source just described is an algebraic direct summand of the cohomology of a product of degree-seven Fermat varieties, with Tate factors supplied by projective spaces.

Shioda's fixed-prime product theorem [12, Theorem 2] gives ordinary HC on all these Fermat products; projective-space factors preserve it by the projective bundle formula. At a very general point, each Hodge class of $A^u \times E^v$ is fixed by M . Semisimplicity of polarizable rational Hodge structures supplies a Hodge preimage under the displayed surjection. The algebraic source projectors and Shioda make that preimage algebraic, and the forward diagram map transports its class to $A^u \times E^v$. Only the forward map, not a chosen Hodge splitting, needs to be algebraic. Taking the countable intersection over u, v leaves one very-general locus on which all these ordinary-Hodge assertions hold. $\square$

Conclusion of Theorem 6.1. Fix $q \geq 1$. The dominating varieties found above are $A^n \times E^t$, and ordinary HC holds for $A^q \times (A^n \times E^t) = A^{q+n} \times E^t$. The domination criterion [13, Proposition 1], attributed there to Grothendieck, proves generalized HC for A^q . This holds for every q . $\square$

Remark 6.2 (Scope and attribution of the corollary). This is an explicit consequence of Rohde's cyclic monodromy theorem, the classical Fermat determinant construction and Shioda's cycle theorem, and Abdulali's domination machinery. No novelty is claimed for the resulting family. The explicit quadratic CM calculation explains why the stronger conjecture follows here: it supplies all the auxiliary fully twisted realizations required by the domination proof. For other primes, realizing one determinant as a Fermat Hodge structure does not by itself identify fully twisted realizations for every irreducible constituent of its tensor algebra. We make no analogous general-prime assertion.

7 All powers for odd-exponent abelian covers

This section extends the prime-exponent results to full Jacobians of abelian covers with odd composite exponent. The cycle-theoretic premise is isolated in a fixed Fermat Jacobian, including all its powers.

Theorem 7.1 (Odd-exponent transfer criterion). *Let $N > 1$ be odd and assume the rational Hodge conjecture for every power of the Jacobian of the degree- N Fermat curve. Let G be a finite abelian group of exponent dividing N , and let $a_1, \dots, a_r \in G \setminus \{0\}$ generate G and have sum zero. Then the rational Hodge conjecture holds for every power of the very general Jacobian in the full ordered branch family of connected G -covers of $\mathbf{P}^1$ with these local monodromies.*

Very general means outside a countable union of proper closed algebraic subsets of $M_{0,r}$. A finite base cover may be used to choose the family and auxiliary algebraic graphs. The case $r = 2$ is a cyclic genus-zero cover; $r = 3$ is included as a constant family. The following subsections prove the theorem and then identify unconditional exponents.

7.1 Primitive quotient factors, without double counting

Fix ζ_N and identify $\widehat{G}$ with $\text{Hom}(G, \mathbf{Z}/N)$. Evaluation on the branch elements gives an injective code $C \subset (\mathbf{Z}/N)^r$; freeness of this module is not assumed. Here N is odd, and the exponent of G divides N . For $c \in C \setminus \{0\}$, let $d = d(c)$ be its additive order and write

$$c_i = (N/d)b_i \quad \text{in } \mathbf{Z}/N, \quad b_i \in \mathbf{Z}/d.$$

The residues b_i generate $\mathbf{Z}/d$, have sum zero, and have the same active support $S(c)$ as c . Set $s(c) = |S(c)|$ and $h(c) = s(c) - 2$. The cyclic quotient $C_c = X/\ker c$ has degree d . Choose its deck generator T_c so that the specified character has eigenvalue ζ_d . Let

$$U_{[c]} = \ker \Phi_d(T_c^* : H^1(C_c, \mathbf{Q}) \longrightarrow H^1(C_c, \mathbf{Q})).$$

Here $[c]$ is the orbit under $(\mathbf{Z}/N)^\times$ and Φ_d is the d th cyclotomic polynomial. The restriction map $(\mathbf{Z}/N)^\times \rightarrow (\mathbf{Z}/d)^\times$ is surjective. Thus

$$U_{[c]} \otimes \mathbf{C} = \bigoplus_{u \in (\mathbf{Z}/d)^\times} V_{uc}, \quad \dim_{\mathbf{Q}(\zeta_d)} U_{[c]} = h(c).$$

Taking one representative per nonzero orbit gives the rational decomposition of $H^1(X, \mathbf{Q})$, omitting the zero-dimensional support-two terms. In particular, we do *not* sum the entire cohomology of all cyclic quotients: that would count their lower-conductor parts repeatedly.

All the displayed primitive projections are rational algebraic. For any d , the projector on primitive eigenvalues of an order- d operator is

$$e_d(T) = \sum_{k|d} \mu(d/k) \frac{k}{d} \sum_{j=0}^{d/k-1} T^{kj}, \quad (4)$$

where μ is the Möbius function. The inner average projects onto eigencharacters whose orders divide k , so Möbius inversion selects order exactly d . For $d = p^f$ this simplifies to $1 - p^{-1} \sum_{j=0}^{p-1} T^{jd/p}$. Composing this polynomial with quotient pullback and norm realizes the primitive factor inside $H^1(X, \mathbf{Q})$. Its polarization and its cyclotomic action are consequently algebraic as well.

7.2 Primitive monodromy at every odd conductor

Let $d > 1$ be odd and let $a = (a_1, \dots, a_s)$ have nonzero entries in $\mathbf{Z}/d$, with $\sum_i a_i = 0$ and $\gcd(d, a_1, \dots, a_s) = 1$. We consider the full ordered branch family of cyclic degree- d covers of $\mathbb{P}^1$ with this word. Put $h = s - 2$. For a unit $b \bmod d$, set

$$Q_a(b) = \frac{1}{d} \sum_i [ba_i]_d, \quad (h_b^{1,0}, h_b^{0,1}) = (s - 1 - Q_a(b), Q_a(b) - 1),$$

where $[x]_d \in \{1, \dots, d-1\}$ for $x \neq 0$. These are the multiplicities of the two weight-one Hodge types, not the bidegrees of a weight- h determinant.

Lemma 7.2 (A good triple outside the small conductors). *Suppose $s \geq 4$ and $d \notin \{3, 5, 15\}$. At every primitive embedding, either the eigenspace is mixed or there are three active labels i, j, k such that the order of $\zeta_d^{a_i+a_j}$ exceeds 5 and at least one of $\zeta_d^{a_i+a_k}, \zeta_d^{a_j+a_k}$ has order exceeding 2. Consequently each primitive complex eigenspace has connected complex monodromy SL_h .*

Proof. If every pair eigenvalue had order at most 5, its odd order would divide 15. Hence $d \mid 15(a_i + a_j)$ for every distinct pair. Three distinct labels give $d \mid 30a_i$ for every i . Since d is odd and the word is primitive, this implies $d \mid 15$, a contradiction. Choose a pair i, j whose eigenvalue has order greater than 5.

If some third label gives a nonzero cross-sum, it gives the desired triple, since every nontrivial root of odd order has order greater than 2. Otherwise $a_i = a_j = a$ and every other entry equals $-a$. For $s \geq 5$, take three of those other entries: their pair sums are $-2a$, of the same order as $2a = a_i + a_j$, so they give a good triple. For $s = 4$, the word is $(a, a, -a, -a)$; its age equals 2 at every primitive embedding and it is mixed.

Orders and the vanishing of cross-sums are unchanged under multiplying the word by a unit. At a mixed embedding $2 \leq Q_a(b) \leq s - 2$. Choose an active label as infinity. Its omission leaves $n = s - 1$ finite weights whose sum is strictly between 1 and $n - 1$, as required by [4, Definition 1.1(a)]. At a definite embedding the age is 1 or $s - 1$; choose a good triple and a different label as infinity. The finite weight sum is respectively less than 1 or greater than $n - 1$, and the triple meets [4, Definition 1.1(b)]. Primitivity persists after omitting the infinity entry, because that entry is the negative sum of the finite ones. Theorem 5.1 of [4] therefore applies. Its real connected group is the full special unitary group; its complexification is SL_h . $\square$

For $d = 3, 5$, the prime-conductor argument of [1, Section 2] applies: every word with $s \geq 4$ has a mixed primitive embedding, and Galois conjugation transports its full complex monodromy to all primitive embeddings. The same Galois argument will be used below: in a cyclotomic basis the monodromy matrices have entries in $\mathbb{Q}(\zeta_d)$, and the Zariski-closure ideal commutes with scalar extension. Applying a field automorphism transports the full SL_h closure.

Lemma 7.3 (The unique all-definite conductor-fifteen word). *For $d = 15$ and $s \geq 4$, all primitive projections have full SL_{s-2} monodromy unless, up to a unit multiple and a permutation, $a = (1, 2, 4, 8)$. This exceptional primitive rational factor is isogenous to E^8 , where E is a fixed elliptic curve with complex multiplication by $\mathbb{Q}(\sqrt{-15})$. Moreover E is an isogeny factor of the degree-fifteen Fermat Jacobian.*

Proof. If some pair eigenvalue has order greater than 5, the preceding good-triple argument applies, including its mixed four-point case. It remains to consider words whose pair sums are all divisible by 3 or 5.

Such a primitive word contains a unit: otherwise it must contain both a nonzero multiple of 3 and a nonzero multiple of 5, and their sum would be a unit. Normalize one unit entry to 1. The other entries then belong to

$$\{2, 4, 5, 8, 9, 11, 14\}.$$

A unit cannot be repeated, since twice a unit is a unit. Directly testing divisibility by 3 or 5 shows that the maximal compatible support sets containing 1 are

$$\{1, 2, 4, 8\}, \quad \{1, 4, 11, 14\}, \quad \{1, 4, 5\}, \quad \{1, 9, 11\}.$$

For example, 5 is compatible only with the units 1, 4 in this list, and 9 only with 1, 11; they are incompatible with each other. Among the units, the first two displayed sets are the maximal cliques. This is a seven-vertex finite divisibility check, not a numerical search over branch configurations.

If all entries are units, $s \geq 4$ forces one of the two four-element sets. The word $(1, 4, 11, 14)$ is two opposite pairs and is mixed. For support involving 5, both units 1, 4 must occur: with only one unit the sum cannot vanish modulo 5. The word therefore is $(1, 4, 5^m)$, with $m = 2 + 3t$ and $t \geq 0$. At embedding $b = 2$, its Hodge multiplicities are $(1 + t, 1 + 2t)$, both positive. For support involving 9, the word similarly is $(1, 11, 9^m)$, $m = 2 + 5t$; at embedding $b = 1$ its multiplicities are $(1 + 2t, 1 + 3t)$. Galois propagation now gives full complex monodromy in all these mixed cases.

For the remaining word $(1, 2, 4, 8)$, put $K = \mathbb{Q}(\zeta_{15})$ and $P = \langle 2 \rangle = \{1, 2, 4, 8\}$. The primitive eigenspaces have multiplicities $(2, 0)$ on P and $(0, 2)$ on $-P$. Write

$$\theta = \zeta_{15} + \zeta_{15}^2 + \zeta_{15}^4 + \zeta_{15}^8.$$

The sum over all primitive fifteenth roots gives $\theta + \bar{\theta} = 1$. The twelve ordered differences of distinct elements of P are exactly the nonzero residues other than 5, 10; hence

$$\theta\bar{\theta} = 4 + \sum_{j \neq 0, 5, 10} \zeta_{15}^j = 4.$$

Thus $\theta^2 - \theta + 4 = 0$, and $K^P = \mathbb{Q}(\sqrt{-15}) =: K_0$.

Choose an elliptic curve E with CM by K_0 and the type whose extensions to K form P . The weight-one rational Hodge structure

$$T = H^1(E, \mathbb{Q}) \otimes_{K_0} K$$

has rank one over K and exactly that pure type at each embedding. The exceptional primitive factor has rank two over K , so any K -linear identification with $T^{\oplus 2}$ is a Hodge isomorphism: each embedding summand has just one weight-one Hodge type. Since $[K : K_0] = 4$, its rational first cohomology is $H^1(E, \mathbb{Q})^{\oplus 8}$. The full faithfulness of rational weight-one Hodge structures for complex abelian varieties up to isogeny gives an actual isogeny with E^8 . In particular this supplies algebraic maps on the entire primitive first cohomology, not merely a numerical determinant-type match. Multiplying the residue word by a unit can exchange the two K_0 types; conjugation on K_0 gives a rational Hodge isomorphism between these realizations after forgetting the specified K_0 action, so the same elliptic isogeny class serves both types. At a fixed fiber these are the algebraic maps required by the invariant-theory assembly; no global choice of an isogeny is needed. The primitive rational summand has a monodromy-stable integral lattice, obtained by intersecting it with integral cohomology. Every real factor of its polarization-preserving cyclotomic centralizer is compact, because every primitive embedding is definite. The monodromy therefore lies in the intersection of a compact real group with the discrete group of integral lattice automorphisms and is finite.

Finally, let C_0 normalize $y^{15} = x(x-1)^2$, whose residues are $(1, 2, 12)$. Its primitive rank-one K -factor has multiplicities $(1, 0)$ on P and $(0, 1)$ on $-P$. Thus its primitive abelian factor is isogenous to E^4 . On the Fermat curve $u^{15} + v^{15} = 1$, the map

$$x = u^{15}, \quad y = uv^2$$

gives a finite covering of C_0 . Pullback and norm therefore make E an isogeny factor of the Fermat Jacobian. All statements about E^8 and E^4 concern the primitive factors: the whole four-branch and three-branch Jacobians have dimensions 14 and 6, respectively. $\square$

7.3 The exact simultaneous blocks at varying conductors

Write M for the connected algebraic monodromy group on the full rational first cohomology. The projection calculations above were made on affine configuration spaces. Normalizing two finite points identifies $\text{Conf}_{s-1}(\mathbf{C})$ with $\text{Aff}_1 \times M_{0,s}$. An affine coordinate change rescales the Kummer variable by a root of a power of its scale factor, so the affine fiber acts by finite deck scalars on every character. Thus each projective character representation descends to $M_{0,s}$ with the full connected projection unchanged. Forgetting inactive labels, $M_{0,r} \rightarrow M_{0,s}$ has connected configuration-space fibers and surjects on fundamental groups. Support-three primitive factors are constant after a finite root choice, by normalizing their three active points to $0, 1, \infty$. Pair-braid determinants and the scalar ambiguities have finite order by [4, Corollary 2.7].

For the indices with $s(c) \geq 4$ other than the finite factors of Lemma 7.3, the complex blocks are exactly those of [1, Section 3], with words now read in $\mathbf{Z}/N$: different supports are independent; on support at least five the block is $\{c, -c\}$; on support four it is

$$(\mathcal{V}_4(S)c \cup -\mathcal{V}_4(S)c) \cap C.$$

We check the changes needed for this extension.

The connected solvable radical of $M_{\mathbf{C}}$ maps to a connected solvable normal subgroup of each full SL_h projection, hence trivially. It also acts trivially on the finite-monodromy factors. Faithfulness therefore makes the radical trivial, and semisimple Goursat theory reduces the product determination to the pairwise graph relations.

All projective character representations are defined on the same $\pi_1(M_{0,r})$, before a finite base cover: two lifts of a marked sphere mapping class differ by a deck transformation, scalar on each character. This asserts a product of projective representations, not one common scalar on their direct sum. In a product of two centerless simple groups the normalizer of a graph of an automorphism is that graph. Hence a graph relation between connected projections constrains the original pair braids, before any powers kill their finite eigenvalues. For an active pair, the rank-one formula of [4, Theorems 2.2 and 2.5], with a different active point at infinity, is nonzero: the n spanning vectors have their sole relation $\sum g_i = 0$, so its vector is nonzero, and the Hermitian form and its coefficient are nondegenerate and nonzero. The projective twist is therefore nonidentity even when the pair sum is zero, in which case it is unipotent. If a label is inactive for the other character, forgetting it kills the other character's pair twist. This proves support separation.

On a fixed support, the exceptional pair eigenvalue is $\zeta_N^{c_i+c_j}$, equivalently $\zeta_{d(c)}^{b_i+b_j}$. In rank $h \geq 3$, its multiplicities $1, h-1$ and the inner-or-dual form of an automorphism of PGL_h give one global sign ϵ and all equations $e_i + e_j = \epsilon(c_i + c_j)$ modulo N . Since 2 is invertible modulo N , three finite indices reconstruct $e_i = \epsilon c_i$; the sum-zero equation reconstructs the infinity coordinate. In rank two, put $x = c_1 + c_2$, $y = c_1 + c_3$, $z = c_1 + c_4$. The inverse

$$c = \frac{1}{2}(x + y + z, x - y - z, -x + y - z, -x - y + z) \quad \text{in } (\mathbf{Z}/N)^4$$

identifies the eight independent sign changes of the three unordered pair ratios with $\{\pm P : P \in \mathcal{V}_4\}$. Zero sums cause only stabilizers. These are necessity arguments over the common ring $\mathbf{Z}/N$; no field assumption is hidden in the reconstruction.

Conversely $e = Pc$ preserves the additive order d of the word. Normalize the four branch points to $0, 1, t, \infty$; the three double transpositions are realized by

$$T_{12|34}(x) = \frac{t(x-1)}{x-t}, \quad T_{13|24}(x) = \frac{x-t}{x-1}, \quad T_{14|23}(x) = \frac{t}{x}.$$

Dividing the two words by N/d gives degree- d Kummer equations $y^d = f_c(x)$ and $z^d = f_e(x)$. The divisor of $f_e(T_P(x))/f_c(x)$ is divisible by d ; a degree-zero divisor on $\mathbf{P}^1$ is principal. Consequently the quotient is $a(t)h(x)^d$, where $a(t)$ is a constant times powers of t and $t-1$. Adjoining $a(t)^{1/d}$ gives the deck-equivariant isomorphism $(x, y) \mapsto (T_P(x), a(t)^{1/d}h(x)y)$ on the smooth normalizations. The finite root cover is etale on $M_{0,4}$. Pullback and norm, followed by (4), give algebraic isomorphisms of their primitive factors at every primitive embedding. Thus $O_c = (\mathcal{V}_4(S)c) \cap C$ is the actual oriented component; $-O_c$ is its polarized dual, possibly equal to O_c . The negative orientation uses polarization as in [1]; it is not silently replaced by a degree-zero Hodge isomorphism. Thus no graph can identify different conductors. Goursat and the faithful standard occurrences give the stated product of SL_h factors, with no additional connected scalar factor or hidden pairwise relations. Multiplication of all character words by a unit modulo N preserves the description, so it is Galois-stable.

7.4 Primitive determinants as Fermat Chow summands

Let C_c have odd degree d , active support $s = h + 2 \geq 3$, and primitive factor $U = U_{[c]}$. Its rational determinant is

$$D_c \otimes \mathbf{C} = \bigoplus_{u \in (\mathbf{Z}/d)^\times} \bigwedge^h V_{uc}, \quad \dim_{\mathbf{Q}} D_c = \varphi(d).$$

It is realized in both $H^h(C_c^h, \mathbf{Q})$ and $H^h(J(C_c), \mathbf{Q})$. The finite-quotient determinant construction at arbitrary degree, including the primitive cyclotomic factor and the deck action in one slot, already appears in Schoen [6, Lemmas 1.1–1.2, pp. 5–6]. We record its primitive projector and the normalization of the two correspondences explicitly, extending the presentation in [3]; the composite-degree mechanism is not claimed as a new construction.

Write $y^d = \prod_{i=1}^{h+2} (x - t_i)^{b_i}$ with all active points finite and $\sum b_i = 0$ modulo d . Put $Z = C_c^h$ and

$$H = \ker(\mu_d^h \xrightarrow{\Pi} \mu_d) \rtimes \mathfrak{S}_h, \quad \tilde{H} = \mu_d^h \rtimes \mathfrak{S}_h.$$

The quotient $Y = Z/H$ has function field

$$\mathbf{C}(\mathrm{Sym}^h \mathbf{P}^1)(z), \quad z^d = \prod_i L_i^{b_i},$$

where L_i is evaluation at t_i . The Kummer order is exactly d because $\gcd(d, b_1, \dots, b_s) = 1$. Likewise the independent coordinate Kummer classes on the ordered product have order d , as their coordinate valuations show. The $h+2$ evaluation hyperplanes are in general position, with a unique relation having all coefficients nonzero. Taking d th roots of these coordinates produces a smooth diagonal Fermat hypersurface F_d^h finite over $\mathbf{P}^h$. The character $(\epsilon_i) \mapsto \prod_i \epsilon_i^{b_i}$ of μ_d^{h+2}/μ_d is surjective, with kernel of order d^h . Its quotient has the same function field as Y ; normality identifies the quotients globally. Thus there are finite maps

$$F_d^h \xrightarrow{r} Y \xleftarrow{q} Z, \quad d_r = d^h, \quad d_q = d^{h-1}h!.$$

Let $e_H, e_{\tilde{H}}$ be the rational averaging correspondences. The group H is the kernel of the product map $\tilde{H} \rightarrow \mu_d$, hence is normal. A deck generator T acting in one coordinate therefore commutes with both averages. The following is a rational Chow idempotent on Z :

$$P_d = e_d(T)(e_H - e_{\tilde{H}}). \quad (5)$$

On cohomology, the kernel-of-product average forces all deck labels to agree. The primitive projector forces that label to be primitive and nonzero, so every factor is in $H^1(C_c)$; H^0 and H^2 have trivial deck label. The unsigned geometric permutation average acts on these degree-one tensors as the ordinary antisymmetrizer, by the Koszul sign. Consequently the image of P_d is concentrated in degree h and is exactly D_c . Lower-conductor eigenspaces of the entire cyclic quotient are removed by $e_d(T)$.

For $\mathcal{T} = [F_d^h \times_Y Z]$, the finite-correspondence identity is

$$\mathcal{T}\mathcal{T}^\dagger = d_r \sum_{g \in H} \Gamma_g.$$

This holds in rational Chow groups. The relevant intersection is proper and finite over the dimension- h base; its generic components are the graphs in the displayed sum. Components supported over the branch locus have dimension less than h and cannot contribute to that cycle. It follows that

$$R_d = (d_r d_q)^{-1} \mathcal{T}^\dagger P_d \mathcal{T}$$

is a rational algebraic idempotent on F_d^h . The maps $P_d \mathcal{T}$ and $(d_r d_q)^{-1} \mathcal{T}^\dagger P_d$ identify its image with the P_d summand in both directions. The Jacobian realization uses the algebraic degree-one projector, the Abel–Jacobi H^1 isomorphism and its inverse, and alternation. Choose any basepoint on C_c . A composite cyclic cover need not have a point fixed by its whole deck group, but changing the basepoint changes Abel–Jacobi by a translation, which acts trivially on H^1 . Thus its H^1 isomorphism intertwines the deck actions without a fixed-point assumption, as observed in [6, Remark 1.4, pp. 6–7]. The inverse weight-one Hodge map is algebraic by the divisor realization of homomorphisms of Jacobians. The addition and diagonal formulas give the same nonzero $h!$ normalization as in [3]; none depends on d being prime. The asserted Chow idempotent identities remain on C_c^h and F_d^h ; the Jacobian comparison here is an algebraic isomorphism of the specified cohomological summands.

7.5 The precise Fermat premise and its established cases

Assume the rational Hodge conjecture on every power of $J(F_N^1)$. We need arbitrary products of Fermat hypersurfaces, and record the reduction instead of silently treating a theorem about curves as a theorem with that different statement. Aoki’s original [15, Theorem 0.1, p. 177] supplies this premise when N is a power of an odd prime, or when $N = 3^a 5^b$ or $N = 3^a 7^b$ with $a, b \geq 0$ and $N > 1$: his assertion covers all abelian varieties isogenous to factors of powers of the corresponding Fermat Jacobian.

First the conjecture on $J(F_N^1)^q$ implies it on $(F_N^1)^q$: Abel–Jacobi pullback surjects on the cohomology of the curve in degrees 0, 1, 2, hence on the product, and semisimplicity lifts rational Hodge classes through this surjection. Pullback of the resulting algebraic classes is algebraic.

Write $F_N^a = \{\sum_{i=0}^{a+1} x_i^N = 0\}$. For $a, b \geq 1$, the Shioda–Katsura construction [16, Lemma 1.2 and Theorem 1.7] resolves the dominant degree- N rational map

$$F_N^a \times F_N^b \dashrightarrow F_N^{a+b}, \quad (x, y) \mapsto [x_0 y_{b+1} : \cdots : x_a y_{b+1} : \eta x_{a+1} y_0 : \cdots : \eta x_{a+1} y_b], \quad \eta^N = -1,$$

by blowing up the smooth codimension-two center $F_N^{a-1} \times F_N^{b-1}$ defined by $x_{a+1} = y_{b+1} = 0$. Indeed the base ideal is locally generated by these two coordinates; the blowup removes their common factor in the displayed sections. Their N th powers sum to zero by the defining equations. Away from the center, recovering the two original factors leaves exactly N choices. Thus the resolved morphism f satisfies $f_* f^* = N \text{id}$ on rational cohomology.

Apply this with $a = h - 1, b = 1$ to a Fermat factor of dimension $h \geq 2$, and take its product with all the other factors. The blowup formula expresses its cohomology algebraically as that of the product with $F_N^{h-1} \times F_N^1$, plus the center $F_N^{h-2} \times F_N^0$ with a Tate twist. Induction on $\sum_j \max(h_j - 1, 0)$, starting with curve products and points, proves the Hodge conjecture for every product $\prod_j F_N^{h_j}$. Here F_N^0 is a disjoint union of N points. At every stage both summand maps are algebraic, and $N^{-1}f_*$ descends algebraic lifts. Finally, if $d \mid N$, the finite power map

$$F_N^h \longrightarrow F_d^h, \quad [x_i] \longmapsto [x_i^{N/d}]$$

gives the same conclusion for arbitrary products with degrees dividing N . The primitive determinant transfers just proved imply that every rational Hodge class in every full rational tensor product of the D_c is algebraic in its specified realization. In addition, if E is the constant elliptic factor supplied by the conductor-fifteen classification, then $H^1(E, \mathbf{Q})$ is an algebraic direct summand of $H^1(F_{15}^1, \mathbf{Q})$, and hence of $H^1(F_N^1, \mathbf{Q})$ whenever $15 \mid N$. An isogeny factor of the Fermat Jacobian gives the two H^1 correspondences by the isogeny projection and its rational inverse, composed with Abel–Jacobi; equality of Hodge numbers alone would not suffice. Therefore the same conclusion holds for mixed tensors of the D_c and any number of copies of $H^1(E, \mathbf{Q})$, after effective Tate twists and rational algebraic projections as well.

7.6 Rational invariant assembly with different conductors

It remains to explain why mixed conductors require no unsupported matching-embedding assertion. Use the full rational sources

$$E_{c_1, \dots, c_t; u} = D_{c_1} \otimes_{\mathbf{Q}} \cdots \otimes_{\mathbf{Q}} D_{c_t} \otimes_{\mathbf{Q}} H^1(E, \mathbf{Q})^{\otimes u},$$

including repeated entries and the empty source $\mathbf{Q}$. Here $u = 0$ unless the exceptional conductor-fifteen factors occur; each such primitive factor is supplied by the classification as an actual weight-one Hodge structure isomorphic to $H^1(E, \mathbf{Q})^{\oplus 8}$. The corresponding abelian varieties are isogenous because rational weight-one Hodge morphisms between complex abelian varieties are induced by homomorphisms up to isogeny. Thus these exceptional slots are inserted by rational algebraic maps. Such a source generally is *not* rank one over $\mathbf{Q}(\zeta_N)$. That property is neither claimed nor needed.

For each of the product- SL_h blocks, tensor invariant theory expresses all invariants through pair contractions and determinant volumes. Actual oriented graphs transfer occurrences to representatives; polarization handles their dual occurrences. This includes the self-dual rank-two blocks and the support-three rank-one factors. The exceptional rank-two primitive factors are constant after finite base change, so their slots are unrestricted tensors of the additional $H^1(E)$ source. A diagram of total degree $2k$ therefore gives an algebraic insertion map from some $E_{c_1, \dots, c_t; u}(-v)$, with $\sum h(c_i) + u + 2v = 2k$ and $v \geq 0$, into $H^{2k}(A^q, \mathbf{Q})$.

For precision about rational spanning, work temporarily over $L = \mathbf{Q}(\zeta_N)$. Every desired primitive character selector on a raw H^1 slot is the Fourier idempotent

$$e_c = \frac{1}{|G|} \sum_{g \in G} \zeta_N^{-c(g)} g^* \quad \text{in } L[G].$$

It is an L -linear combination of rational algebraic deck correspondences. Each required determinant line is a component of its full rational D_c . A contraction is obtained by applying the two character selectors to the rational polarization tensor, and an oriented identification by applying selectors to its actual rational graph correspondence. Alternation and passage to exterior cohomology are

rational algebraic, including their factorial normalizations. Thus, on expanding these finitely many Fourier selectors, every L -valued invariant diagram belongs to the L -linear span of images of rational algebraic maps from the full sources $E(-v)$ above. All these maps intertwine connected monodromy; the determinant sources have finite determinant monodromy and the additional elliptic sources map to constant primitive factors. Their images therefore lie in the invariant part. There are only finitely many diagrams, deck terms and source choices in a fixed cohomological degree. The resulting finite sum is a rational Hodge surjection

$$\bigoplus_{\alpha} E_{\alpha}(-v_{\alpha}) \rightarrow H^{2k}(A^q, \mathbf{Q})^M.$$

Surjectivity follows after extension to L from the spanning just proved, hence holds over $\mathbf{Q}$. This argument also supplies the coefficients needed to distinguish conjugate contraction slots; it does not merely average one chosen diagram or assume a rational monomial basis. Equivalently, one may refine the sources by the rational Galois-orbit idempotents in their tensor products of cyclotomic deck algebras, but that refinement is unnecessary.

For a very general fiber, connected monodromy is contained in its Hodge group, so every rational Hodge class lies in the displayed target. Polarizable rational Hodge structures are semisimple; a Hodge class therefore lifts to a rational Hodge class in the source. The preceding Fermat argument makes every such source class algebraic, and the insertion maps preserve algebraicity. This proves the required surjectivity of the cycle class map in every degree and on every power.

Finally the very-general assertion uses exactly the monodromy–Hodge normality and algebraic Hodge-locus argument in [1, Section 6], which has no prime-exponent hypothesis. Apply the Hodge-locus theorem after the appropriate weight-zero Tate twist and take the countable union over powers and degrees. Images of these proper closed loci under the finite base map remain proper and closed. The conclusion descends to the original full branch family. Together with the simultaneous-block and exceptional-constant classification proved above, this completes Theorem 7.1.

Corollary 7.4 (Aoki exponents). *The conclusion of Theorem 7.1 holds unconditionally on its stated very general locus if*

$$N = p^e \quad (p \text{ odd prime, } e \geq 1), \quad N = 3^a 5^b, \quad \text{or} \quad N = 3^a 7^b,$$

where $a, b \geq 0$ and $N > 1$.

Proof. Aoki’s [15, Theorem 0.1(ii)] applies to every odd prime power. Theorem 0.1(i), for $m = 2^a 3^b 5^c 7^d$ with $c = 0$ or $d = 0$, gives the other two odd families. His definition of Fermat type includes every isogeny factor of every power of the Fermat Jacobian, so it supplies precisely the hypothesis of the criterion. $\square$

8 Verification and prior-art boundary

The exceptional-space arithmetic was independently checked for all odd subgroups of prime residue groups with $p \leq 103$, with zero through three added opposite pairs and $s \geq 5$: 152 cases. The checks verify branch balance, signature rows, generator inversion, diagonal injectivity and inverse-parameter duplicates. A further 7,746 volume words of lengths two and four at $p = 7, 13, 19$ verify the opposite-coset balance criterion. The odd-exponent regressions check eleven exact rational primitive projectors, six finite-abelian character inventories, and 24,159 primitive support-four words at odd conductors at most 45, together with the conductor-fifteen CM type and a cover

without a deck-fixed basepoint. Normal and optimized Python give identical reports. They do not verify algebraic cycles, deformation, monodromy, or the universal quantifiers proved above.

The determinant-to-Fermat construction has direct precedent in Schoen’s 1989 work, as documented by [3]; the algebraicity input there is Shioda’s theorem on products of prime-degree Fermat varieties. The alternative simple-tuple cycle construction is Schoen’s [6, 7], extended by the standalone fusion proof. Although the moving Jacobians in Section 3 are non-CM, their determinant motives are direct summands of Fermat motives. Their exceptional-cycle algebraicity is therefore a consequence of this classical construction and Shioda, not a new algebraicity theorem obtained by observing that the entire Jacobian is non-CM. The three-point boundary is itself a Fermat quotient and is excluded from the moving simple assertion; see also the prime-Fermat results summarized in Aoki [8]. Patel–Zhang [9] extend Schoen’s construction for *etale* abelian covers and explicitly leave ramified cases to future work; an exact comparison of alternative realizations remains necessary. Rohde [10, Theorem 5.3.7] already gives the full prime-cyclic monodromy input, making the single-cyclic ordinary all-powers consequence a deduction from published group and cycle results. Spelta–Tamborini [11, Theorem 1.7 and Section 3] treat simultaneous abelian-cover monodromy under a nonrepetition condition and analyze examples with repeated factors; their results are direct precedent for the independence question. The signature-orbit mechanism in Section 2 is standard and appears explicitly in Xue–Zarhin [5]. The degree-seven generalized-Hodge corollary is attributed in Section 6, including its extra CM and domination steps. The odd-exponent extension in Section 7 uses Menet–Nguyen’s good-weight criterion and an explicit treatment of the constant conductor-fifteen exception. Its cycle transfer uses the classical Shioda–Katsura construction and Aoki’s all-powers Fermat theorem. Schoen’s [6, Lemmas 1.1–1.2 and Remark 1.4] already supplies the arbitrary-degree primitive determinant mechanism and the Abel–Jacobi translation observation. The small residue computations are not premises of this extension, and the extension makes no all-odd-exponent claim without its stated Fermat premise. None of these comparisons constitutes a priority claim for the statements of this paper.

All results depend on the companion geometric proofs and their published inputs. The existing Lean files formalize only designated arithmetic and abstract interfaces. Independent specialist review of the mathematical manuscripts remains necessary. The generalized assertion has the additional published domination input and the explicit quadratic CM verification; ordinary all-powers algebraicity alone is not substituted for them.

References

- [1] K. Kistner, *All powers of very general Jacobians of elementary abelian covers*, 5 September 2026. Companion source `manuscripts/general_all_powers.tex`.
- [2] K. Kistner, *Balanced mixed determinants of prime cyclic covers: a revised fusion argument*, 4 September 2026. Companion source `manuscripts/fusion_revised.tex`, Theorem 1.1 and Sections 2–6.
- [3] K. Kistner, *Determinants of cyclic covers of the line: a finite-quotient reduction to Fermat varieties*, 5 September 2026. Companion source `manuscripts/fermat_determinant_transfer.tex`, Theorem 1.1 and Corollary 1.2.
- [4] G. Menet and D.-M. Nguyen, *Representations of braid groups via cyclic covers of the sphere: Zariski closure and arithmeticity*, arXiv:2310.10401v3, <https://arxiv.org/html/2310.10401v3>.

- [5] J. Xue and Y. G. Zarhin, *Centers of Hodge groups of superelliptic Jacobians*, preprint, version 4, arXiv:0907.1563, Lemma 3.9, Theorem 3.11 and Section 3.13.
- [6] C. Schoen, *Hodge classes on self-products of a variety with an automorphism*, Compositio Math. 65 (1988), 3–32, https://www.numdam.org/item/CM_1988__65_1_3_0.pdf.
- [7] C. Schoen, *Addendum to: Hodge classes on self-products of a variety with an automorphism*, Compositio Math. 114 (1998), 329–336, <https://doi.org/10.1023/A:1000566205021>.
- [8] N. Aoki, *Hodge cycles on CM abelian varieties of Fermat type*, Comment. Math. Univ. St. Pauli 51 (2002), https://rikkyo.repo.nii.ac.jp/record/8744/files/AA00610867_51-01_07.pdf.
- [9] D. Patel and Y. Zhang, *Algebraicity of Hodge classes on some generalized Prym varieties*, arXiv:2506.13729v2, <https://arxiv.org/html/2506.13729v2>.
- [10] J. C. Rohde, *Cyclic coverings, Calabi–Yau manifolds and Complex multiplication*, dissertation, Universität Duisburg–Essen, 2007, Construction 3.2.1 and Theorem 5.3.7 (printed p. 64), <https://webdoc.sub.gwdg.de/ebook/dissts/Duisburg/Rohde2007.pdf>.
- [11] I. Spelta and C. Tamborini, *Decomposable abelian G -curves and special subvarieties*, Bull. Sci. Math. 201 (2025), 103616, doi:10.1016/j.bulsci.2025.103616; <https://arxiv.org/pdf/2305.19030v2>.
- [12] T. Shioda, *The Hodge conjecture and the Tate conjecture for Fermat varieties*, Proc. Japan Acad. Ser. A Math. Sci. 55 (1979), 111–114, Theorem 2, p. 112, doi:10.3792/pjaa.55.111.
- [13] S. Abdulali, *Tate twists of Hodge structures arising from abelian varieties of type IV*, J. Pure Appl. Algebra 216 (2012), 1164–1170, Proposition 1, Lemma 9, Theorems 10–11 and Remark 12, doi:10.1016/j.jpaa.2011.10.005; <https://arxiv.org/html/1203.4857v1>.
- [14] S. Abdulali, *Tate twists of Hodge structures arising from abelian varieties*, in Recent Advances in Hodge Theory: Period Domains, Algebraic Cycles, and Arithmetic, London Math. Soc. Lecture Note Ser. 427, Cambridge Univ. Press, 2016, 292–307, Appendix B.5, <https://myweb.ecu.edu/abdulalis/Vancouver.pdf>.
- [15] N. Aoki, *Some remarks on the Hodge conjecture for abelian varieties of Fermat type*, Commentarii Mathematici Universitatis Sancti Pauli **49** (2000), 177–194, especially Theorem 0.1(i)–(ii), p. 177. https://rikkyo.repo.nii.ac.jp/record/9851/files/AA00610867_49-02_04.pdf.
- [16] T. Shioda and T. Katsura, *On Fermat varieties*, Tohoku Mathematical Journal **31** (1979), 97–115, especially Lemma 1.2 and Theorem 1.7. https://www.jstage.jst.go.jp/article/tmj1949/31/1/31_1_97/_pdf/-char/en.